Kitaev chain: the physics bridge

From lattice Hamiltonian to Majorana edge zero mode

- original lattice Hamiltonian $\to$ Majorana representation
- Fourier transform $\to$ BdG Hamiltonian
- Dirac / two-band form $\to$ winding number
- continuum domain-wall picture $\to$ edge zero mode

Revised seminar deck | Based on Herviou, *Topological Phases and Majorana Fermions* (2017)

1. Starting point: the lattice Kitaev Hamiltonian

$$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta \sum_j \left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j \right) \quad (1)$$

- spinless fermions, nearest-neighbor hopping, and p-wave pairing
- for the rest of the talk we choose $\Delta \in \mathbb{R}$
- the question is: how do topology and edge zero modes emerge from this lattice model?

Plan of attack

- rewrite in Majorana operators
- diagonalize the bulk in momentum space
- identify the winding of the BdG vector
- pass to a continuum Dirac description near the transition

2. Majorana representation with the essential derivation

Define

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{2}$$

so that

$$c_j = \frac{\gamma_{A,j} + i\gamma_{B,j}}{\sqrt{2}}, \qquad c_j^\dagger = \frac{\gamma_{A,j} - i\gamma_{B,j}}{\sqrt{2}}. \tag{3}$$

The onsite term becomes

$$c_j^\dagger c_j = \frac{1}{2} + i\gamma_{A,j}\gamma_{B,j}. \tag{4}$$

For one bond, substituting c_j, c_{j+1} gives

$$\begin{aligned} &-t\left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j\right) + \Delta\left(c_j^\dagger c_{j+1}^\dagger + c_{j+1} c_j\right) \\ &\quad = i(\Delta + t)\gamma_{B,j+1}\gamma_{A,j} + i(\Delta - t)\gamma_{A,j+1}\gamma_{B,j}. \end{aligned} \tag{5}$$

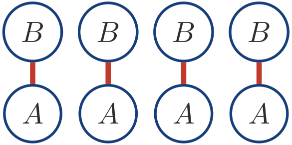
Therefore

$$\begin{aligned} H_K = &-\mu \sum_j \left(i\gamma_{A,j}\gamma_{B,j} + \frac{1}{2}\right) \\ &+ i\sum_j \Big[(\Delta + t)\gamma_{B,j+1}\gamma_{A,j} \\ &\qquad + (\Delta - t)\gamma_{A,j+1}\gamma_{B,j}\Big]. \end{aligned} \tag{6}$$

At $\mu = 0,\ t = \Delta$, only the bond coupling $i\gamma_{B,j+1}\gamma_{A,j}$ survives. Open boundaries therefore leave one unpaired Majorana at each end.

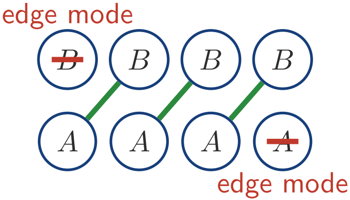
3. Schematic: trivial pairing vs topological pairing

Trivial regime



Dominant onsite pairing $i\gamma_{A,j}\gamma_{B,j}$. No protected boundary mode.

Topological regime: $\mu = 0,\ t = \Delta$



Bond pairing $i\gamma_{A,j}\gamma_{B,j+1}$ leaves one unpaired Majorana at each end.

$$\mu = 0,\ t = \Delta \quad \Longrightarrow \quad H_K = 2it \sum_{j=1}^{L-1} \gamma_{B,j+1}\gamma_{A,j}. \tag{7}$$

4. From real space to BdG: Fourier transform and Nambu spinor

Take periodic boundary conditions and define

$$c_j = \frac{1}{\sqrt{L}} \sum_{k\in \mathrm{BZ}} c_k e^{ikj}, \; c_j^\dagger = \frac{1}{\sqrt{L}} \sum_{k\in \mathrm{BZ}} c_k^\dagger e^{-ikj}. \quad (8)$$

Then

$$H_K = \sum_{k\in \mathrm{BZ}} \xi_k c_k^\dagger c_k + i\Delta \sin k \left(c_k^\dagger c_{-k}^\dagger - c_{-k} c_k \right), \quad (9)$$

with

$$\xi_k = -\mu - 2t \cos k. \quad (10)$$

Introduce the Nambu spinor

$$\Psi_k^\dagger = \left(c_k^\dagger, \; c_{-k} \right). \quad (11)$$

Then the Hamiltonian becomes a sum of 2×2 BdG blocks:

$$H_K = \sum_{k>0} \Psi_k^\dagger h(k) \Psi_k \quad (12)$$

with

$$h(k) = \begin{pmatrix} \xi_k & 2i\Delta \sin k \\ -2i\Delta \sin k & -\xi_k \end{pmatrix}. \quad (13)$$

5. BdG spectrum and the Dirac / two-band form

The spectrum is

$$E_k = \sqrt{(-\mu - 2t\cos k)^2 + (2\Delta\sin k)^2}. \tag{14}$$

Hence the gap closes at

$$\mu = \pm 2t. \tag{15}$$

Now rewrite the BdG matrix as a Dirac / two-band Hamiltonian

$$h(k) = n_y(k)\sigma^y + n_z(k)\sigma^z, \tag{16}$$

with

$$n_y(k) = -2\Delta\sin k, \qquad n_z(k) = -\mu - 2t\cos k.$$

This is the key geometric object. As k runs through the Brillouin zone, the vector

$$\boldsymbol{n}(k) = (n_z(k), n_y(k)) \tag{17}$$

traces out a closed curve in the (n_z, n_y) plane. The phase is determined by whether that curve winds around the origin.

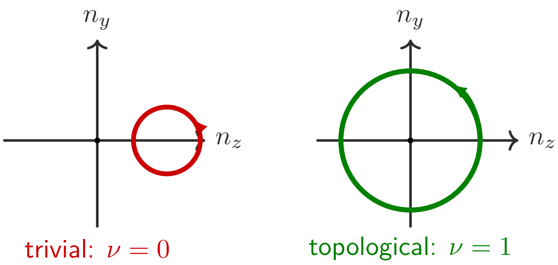
6. Winding number: does the closed curve wrap the origin?

Define

$$\theta_k = \arg\big(n_z(k) + i n_y(k)\big). \tag{18}$$

Then the winding invariant is

$$\nu = \frac{1}{2\pi} \int_{\mathrm{BZ}} \partial_k \theta_k \, dk. \tag{19}$$



Interpretation:

- $\nu = 0$: the loop does *not* enclose the origin
- $|\nu| = 1$: the loop encloses the origin once
- the only way to change ν is for the loop to pass through the origin, i.e. for the bulk gap to close

8. Infinite-chain spectrum

Take an infinite translationally invariant chain with $t = \Delta = 1$. Then

$$E_{\pm}(k) = \pm\sqrt{(-\mu - 2\cos k)^2 + 4\sin^2 k}. \tag{20}$$

Representative cases:

- $\mu = -3$: trivial, gapped
- $\mu = -2$: critical, gap closes at $k = 0$
- $\mu = -1$: topological, gapped